Exercise 3.5

1.(i) $\frac{3x^3z}{5y^3}$ (ii) $p+4$ (iii) $\frac{3t^2}{4}$ 2.(i) $\frac{3x-4y}{2x-5}$ (ii) $\frac{x^2+xy+y^2}{3(x+2y)}$

3.(i) -5 (ii) $\frac{b-4}{b+2}$ (iii) $\frac{3y}{x-3}$ (iv) $\frac{4(2t-1)}{3}$ 4. $\frac{4}{9}$ 5. x^2+4x+4

Exercise 3.6

1.(i) $\frac{2x}{x-2}$ (ii) $\frac{2x^2+2x-7}{(x+3)(x-2)}$ (iii) x^2+xy+y^2 2.(i) $\frac{2(x-2)}{x-4}$ (ii) $\frac{1-x}{1+x}$

3. $\frac{2x^3+1}{(x^2+2)^2}$ 4. $\frac{x+1}{x^2-2x+4}$ 5. $\frac{(4x^2-1)}{2(4x^2+1)}$ 7. 2 hrs 24 minutes 8. 30 kgs, 20 kgs

Exercise 3.7

1.(i) $2\left|\frac{y^4z^6}{x^2}\right|$ (ii) $4\left|\frac{\sqrt{7}x+\sqrt{2}}{4x-1}\right|$ (iii) $\frac{11}{9}\left|\frac{(a+b)^4(x+y)^4}{(a-b)^6}\right|$ 2.(i) $|2x+5|$

(ii) $|3x-4y+5z|$ (iii) $|(x-2)(7x+1)(4x-1)|$ (iv) $\frac{1}{6}|(4x+3)(3x+2)(x+2)|$

Exercise 3.8

1. (i) $|x^2-6x+3|$ (ii) $|2x^2-7x-3|$ (iii) $|4x^2+1|$ (iv) $|11x^2-9x-12|$

2.(i) 49, -42 (ii) 144, 264 3.(i) 30, 9 (ii) 24, -32

Exercise 3.9

1.(i) $x^2+9x+20=0$ (ii) $3x^2-5x+12=0$ (iii) $2x^2+3x-2=0$

(iv) $x^2+(2-a)^2x+(a+5)^2=0$ 2.(i) $-3, -28$ (ii) $-3, 0$ (iii) $-\frac{1}{3}, -\frac{10}{3}$ (iv) $\frac{1}{3}, \frac{-4}{3}$

Exercise 3.10

1.(i) $-\frac{1}{4}, 2$ (ii) $-2, \frac{9}{2}$ (iii) $-2, 9$ (iv) $-\sqrt{2}, \frac{-5}{\sqrt{2}}$ (v) $\frac{1}{4}, \frac{1}{4}$ 2. 6

Exercise 3.11

1.(i) $\frac{2}{3}, \frac{2}{3}$ (ii) $-1, 3$ 2.(i) $2, \frac{1}{2}$ (ii) $\frac{3+\sqrt{3}}{\sqrt{2}}, \frac{3-\sqrt{3}}{\sqrt{2}}$

(iii) $-1, \frac{23}{3}$ (iv) $\frac{a+b}{6}, \frac{a-b}{6}$ 3. 3.75 seconds

Exercise 3.12

1. $5, -\frac{1}{5}$ 2. 1.5 m 3. 45 km/hr 4. 20 years, 10 years

5.Yes, 12 m, 16 m 6. 72 7. 28 m, 42 m 8. 2 m 9. 7 cm

Exercise 3.13

1.(i) Real and unequal (ii) Real and unequal (iii) Not real

(iv) Real and equal (v) Real and equal 2.(i) 2, 3 (ii) $1, \frac{1}{9}$

Exercise 3.14

1.(i) $\frac{(\alpha+\beta)^2-2\alpha\beta}{3\alpha\beta}$ (ii) $\frac{\alpha+\beta}{(\alpha\beta)^2}$ (iii) $9\alpha\beta-3(\alpha+\beta)+1$

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